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AI School Iteration 2.6

Random Forests, Naive Bayes and SVM

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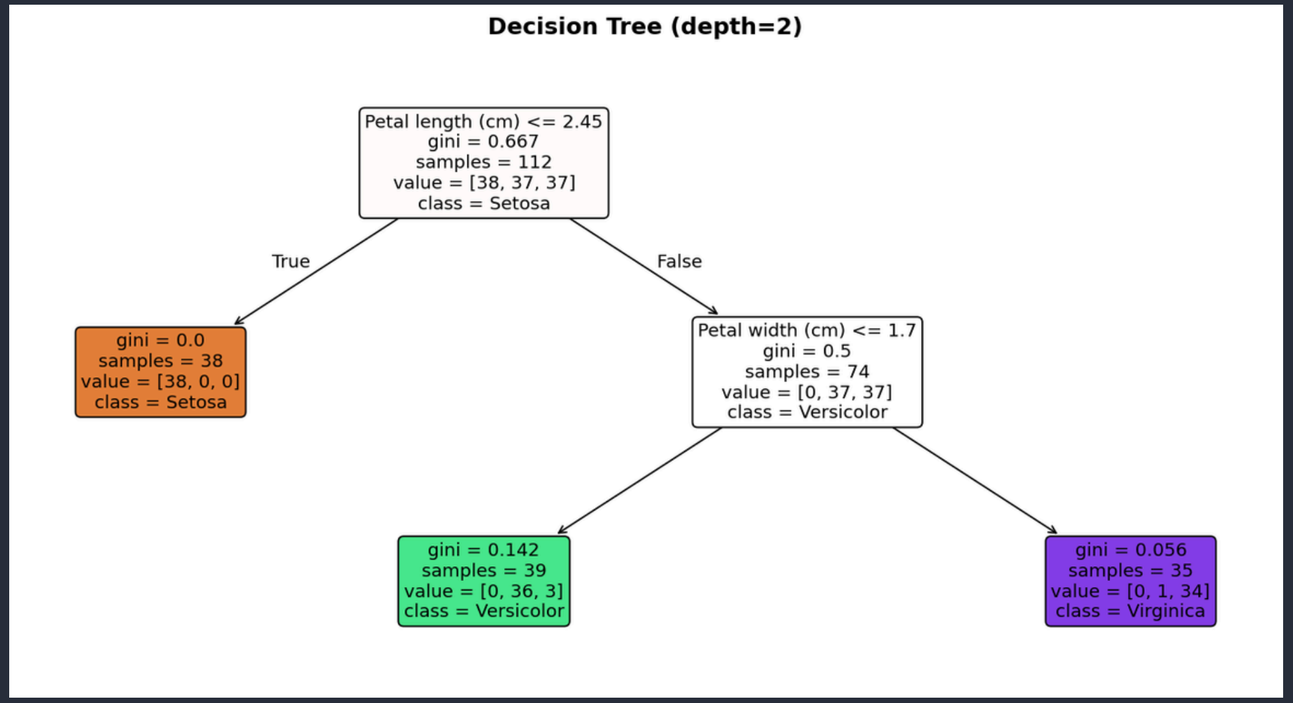
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01. Decision Tree Recap

A Decision Tree learns a **hierarchy of binary rules** on the features.
- Each **internal node** = a split condition: feature ≤ threshold?
- Each **leaf** = a class prediction



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02. The Problem with a Single Tree

Decision Trees are unstable

> Change a few training points → completely different tree.

	Single Decision Tree
Accuracy	**Good**
Interpretability	**High**
Stability	**Low** (small data changes → big boundary changes)

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The question: Can we keep the accuracy but reduce the instability?

→ **Yes.** *Train many trees and let them vote.*

THIS IS ENSEMBLE LEARNING.

03. Bagging

Step 1: Give each tree different training data



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03. Bagging

Step 1: Give each tree different training data

- For each tree, draw a **random sample with replacement** from the training set
- Each sample is roughly the same size, but some points appear twice, others not at all
- ~37% of points are left out of each sample → called **Out-of-Bag (OOB)** samples, free validation, no separate val set needed

Key idea: different data → different trees → genuine diversity

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04. Feature Randomness

Step 2: Each split considers only a random subset of features

At every split, each tree only considers a **random subset** of features (default: √d).
Stops all trees from always splitting on the same feature → trees are less correlated → averaging works better.

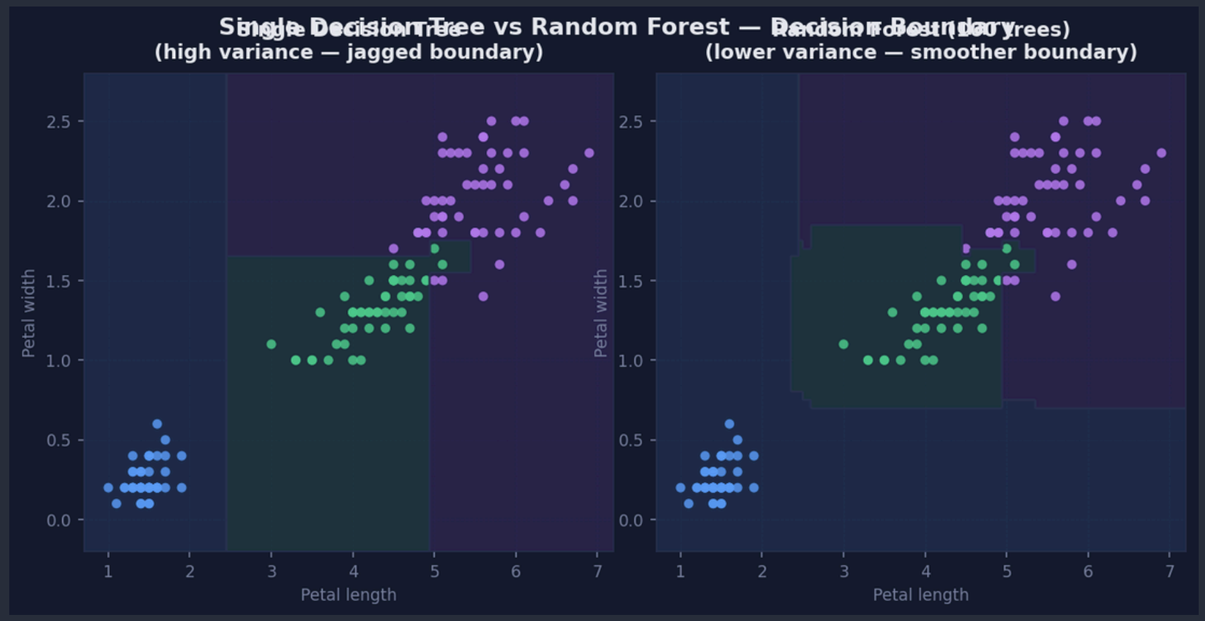
Bagging + feature randomness = a forest of genuinely different trees

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05. Why Averaging Works

One tree makes errors. 100 trees rarely all make the same error in the same place.

=> Averaging reduces mistakes



06. Random Forest in Practice

```
from sklearn.ensemble import RandomForestClassifier


rf = RandomForestClassifier(n_estimators=100, max_features='sqrt',
                            random_state=42)
rf.fit(X_tr, y_tr)
print(rf.feature_importances_) # averaged → stable
```

Scaling not required.

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Naive Bayes

07. Two Approaches to Classification

Discriminative	Generative
Discriminative classifiers learn the boundary between classes (P(y|x)), focusing on direct mapping for classification	generative classifiers learn the underlying distribution of each class (P(x,y) or P(x|y) understand how data is generated
Logistic Regression	Naive Bayes

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08. Bayes' Theorem

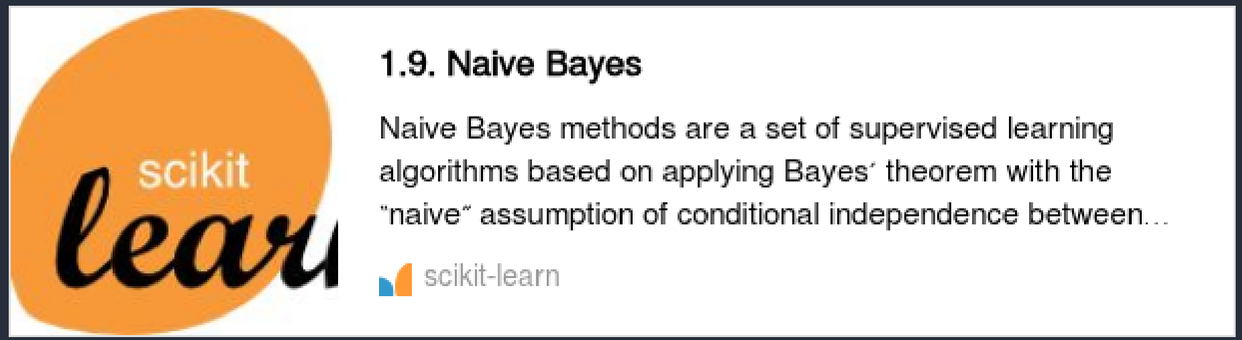
$$P(\text{class} \mid \text{feature}) \propto P(\text{feature} \mid \text{class}) \times P(\text{class})$$

P(class | feature): What we want, the class probability
P(feature | class): How likely is this data in that class?
P(class): How common is this class overall?

Pick the class with the highest product. No training loop needed.

09. The Naive Assumption

Each feature is **independent** of the others, given the class. Almost never true in practice, yet NB often works well. Even when individual probabilities are off, the ranking of classes is frequently correct.



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SUPPORT VECTOR MACHINES (SVM)

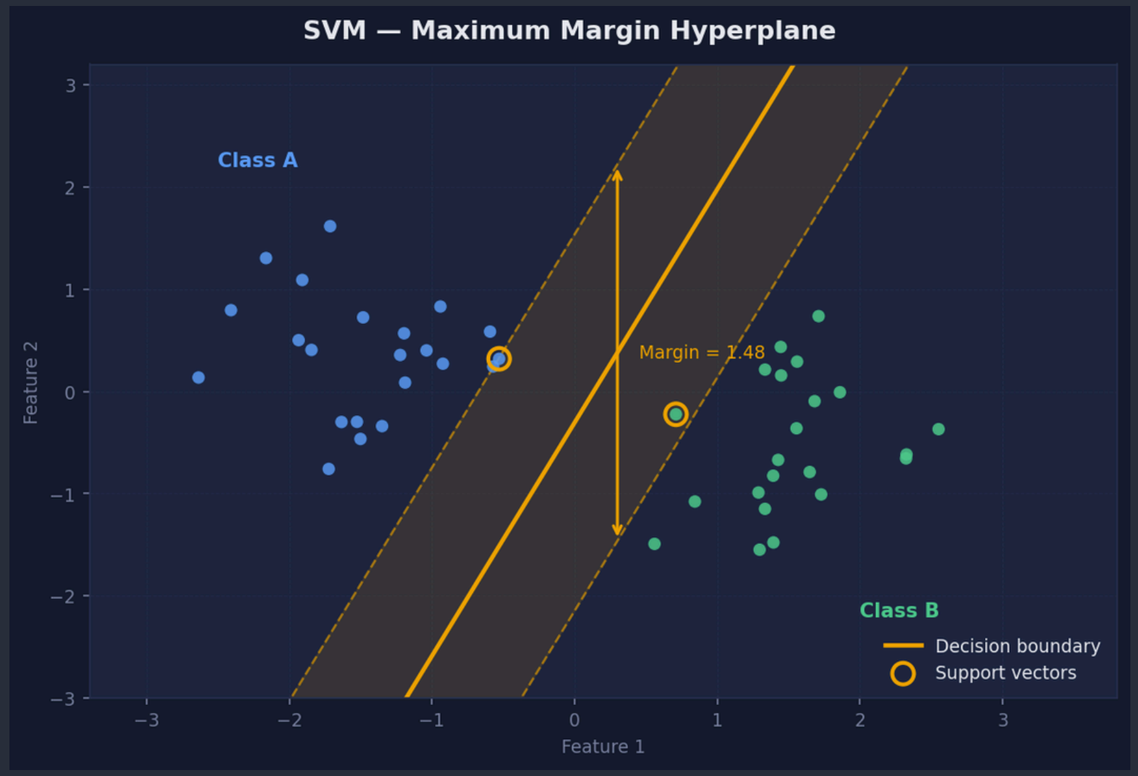
10. The Problem: Too Many Valid Boundaries

> For any linearly separable dataset, infinitely many lines separate the classes.
> Logistic regression picks one. Which one is actually best?

Which boundary is best? → *SVM's answer: the one that is* **as far as possible** *from both classes.*

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11. The Maximum Margin



circled points = support vectors, the only ones that define the boundary

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12. Support Vectors

The boundary is defined by just a few points sitting on the margin edge.

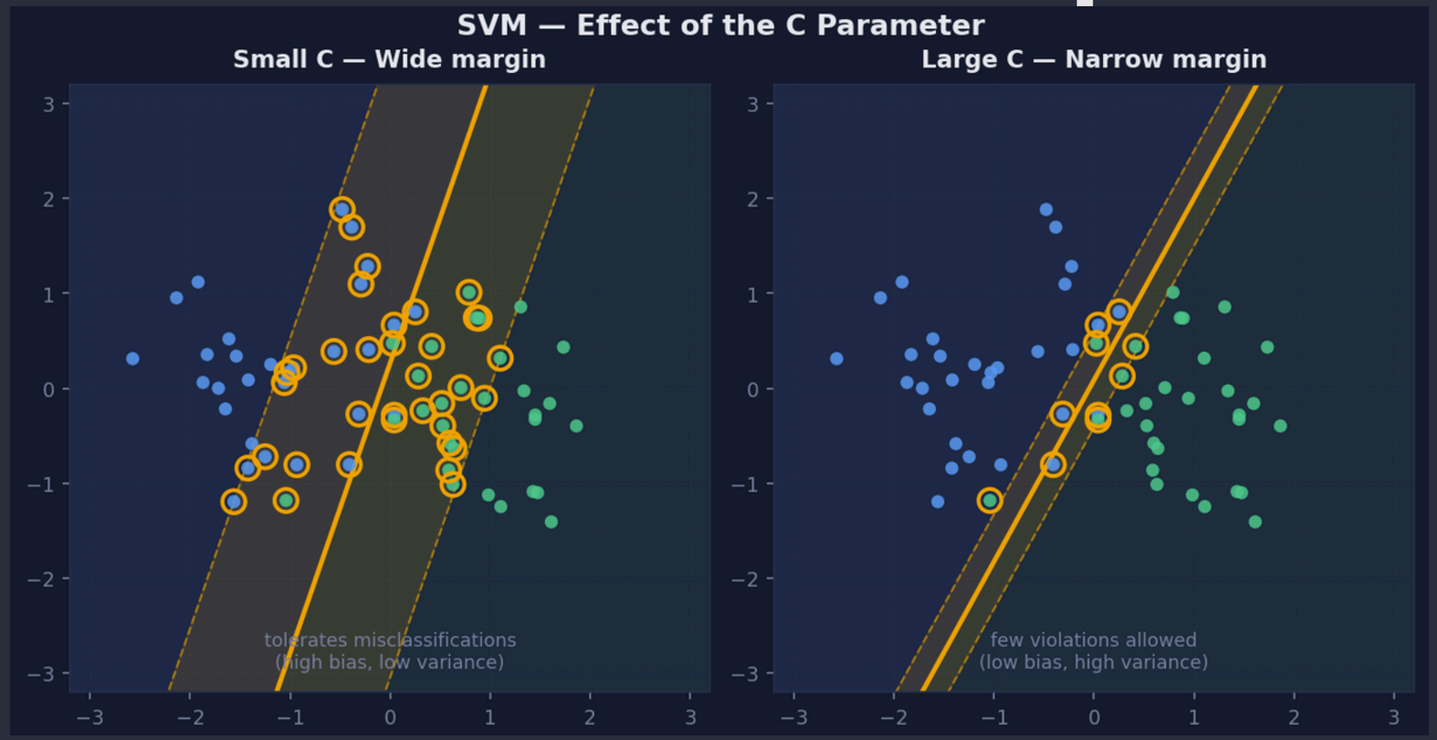
- All other training points are irrelevant, removing them doesn't change the model
- Once trained, only support vectors need to be stored
- Compared to: KNN uses all training points at prediction time

13. The C Parameter

1.4. Support Vector Machines

Support vector machines (SVMs) are a set of supervised learning methods used for classification, regression and outliers detection. The advantages of support vector machines are: Effective in high ...

scikit-learn

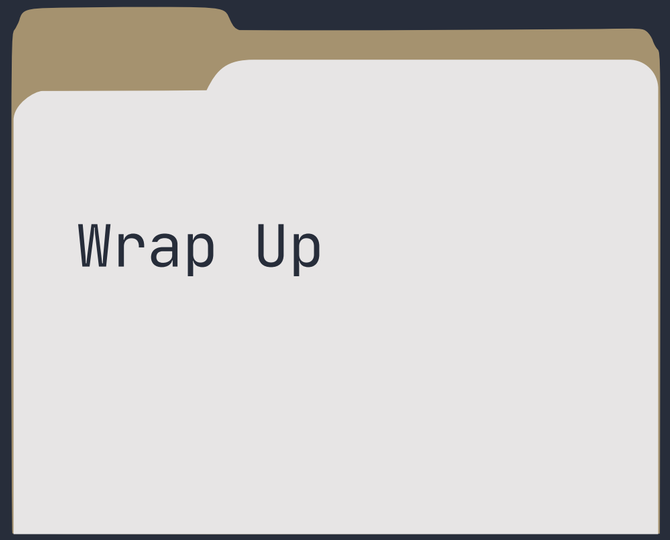


Tolerates misclassifications vs. Tighter fit to training data

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All Five Models

Model	Scaling	Interpretable	Speed
KNN	**Yes**	**No**	Slow at inference
Decision Tree	**No**	**Yes**	Fast
Random Forest	**No**	**Partial**	Moderate
Naive Bayes	**No**	**Partial**	Very fast
SVM	**Yes**	**No**	Slow for large n

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Model Family Tree

Classification models

- Instance-based → KNN
- Rule-based → Decision Tree, Random Forest
- Probabilistic → Naive Bayes
- Margin-based → SVM

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Any Questions ?

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Thank you for your attention ✨

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